

Publications

Total publications: 48 / refereed: 33 / first-author: 5 / selected significant contributions: 7 / collaboration papers: 36 / total citations: 6,313 / h-index: 19 (Last updated: 2026-07-24)

First-author publications (citations: 148)

- 5 **Anand, Abhijeet**; Aguilar, J.; Ahlen, S.; Bianchi, D.; Brodzeller, A.; *et al.*, 2025, *The Cosmic Evolution of C IV Absorbers at $1.4 < z < 4.5$: Insights from 100,000 Systems in DESI Quasars*, The Astrophysical Journal, **990**, 151 (arXiv:2504.20299) [6 citations]
- 4 **Anand, Abhijeet**; Guy, Julien; Bailey, Stephen; Moustakas, John; Aguilar, J.; *et al.*, 2024, *Archetype-based Redshift Estimation for the Dark Energy Spectroscopic Instrument Survey*, The Astronomical Journal, **168**, 124 (arXiv:2405.19288) [44 citations]
- 3 **Anand, Abhijeet**; Kauffmann, Guinevere; & Nelson, Dylan, 2022, *Cool circumgalactic gas in galaxy clusters: connecting the DESI legacy imaging survey and SDSS DR16 Mg II absorbers*, Monthly Notices of the Royal Astronomical Society, **513**, 3210 (arXiv:2201.07811) [30 citations]
- 2 **Anand, Abhijeet**; Nelson, Dylan; & Kauffmann, Guinevere, 2021, *Characterizing the abundance, properties, and kinematics of the cool circumgalactic medium of galaxies in absorption with SDSS DR16*, Monthly Notices of the Royal Astronomical Society, **504**, 65 (arXiv:2103.15842) [65 citations]
- 1 **Anand, Abhijeet**; Roy, Nirupam; & Gopal-Krishna, 2019, *Search for H I emission from superdisk candidates associated with radio galaxies*, Research in Astronomy and Astrophysics, **19**, 083 (arXiv:1812.06875) [3 citations]

Manuscripts in preparation.

- 1 Tan, Joanne; **Anand, Abhijeet**; Naab, Thorsten; *et al.*, 2026, Metal absorption lines and their evolution since $z = 5$ in the IllustrisTNG simulations
- 3 **Anand, Abhijeet**; *et al.*, 2026, Tracing the Cosmic Metal Cycle with DESI: A Uniform Multi-Ion Survey of Intergalactic and Circumgalactic Absorbers

Selected significant contributions (citations: 90)

- 7 Napolitano, Lucas; Myers, Adam D.; Tedeschi, Adam; **Anand, Abhijeet**; Herrera-Alcantar, Hiram K.; *et al.*, 2025, *The Composite Spectrum of QSO Absorption Line Systems in DESI DR2*, ArXiv, arXiv:2512.02992 (arXiv:2512.02992) [1 citations]

My role: Contributed to absorber selection for the stacking analysis and served as an internal reviewer during the collaboration-wide review.

- 6 Lan, Ting-Wen; Prochaska, J. Xavier; Aguilar, J.; Ahlen, S.; **Anand, Abhijeet**; *et al.*, 2025, *The Multi-Phase Circumgalactic Medium of DESI Emission-Line Galaxies at $z = 1.5$* , ArXiv, arXiv:2511.03195 (arXiv:2511.03195) [3 citations]

My role: Contributed to the column density measurements of C IV absorbers identified in DESI quasar spectra and to the interpretation of the results in the context of previous work.

- 5 Wu, Xuanyi; Cai, Z.; Lan, T.-W.; Zou, S.; **Anand, Abhijeet**; *et al.*, 2025, *Tracing the Evolution of the Cool Gas in CGM and IGM Environments through Mg II Absorption from Redshift $z = 0.75$ to $z = 1.65$ Using DESI-Y1 Data*, The Astrophysical Journal, **983**, 186 (arXiv:2407.17809) [12 citations]

My role: Contributed to the discussion section and to the interpretation of the results in the context of the circumgalactic medium of emission-line galaxies.

- ⁴ Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Napolitano, Lucas; **Anand, Abhijeet**; et al., 2024, *Probing the Impact of Radio-mode Feedback on the Properties of the Cool Circumgalactic Medium*, The Astrophysical Journal, **974**, 191 (arXiv:2405.08314) [9 citations]

My role: Contributed to writing and to placing the results in the context of previous studies of gas profiles around luminous red galaxies.

- ³ Galiullin, Ilkham; Rodriguez, Antonio C.; El-Badry, Kareem; Szkody, Paula; **Anand, Abhijeet**; et al., 2024, *Searching for new cataclysmic variables in the Chandra Source Catalog*, Astronomy and Astrophysics, **690**, A374 (arXiv:2408.00078) [7 citations]

My role: Contributed to cross-matching the Chandra Source Catalog with SDSS and DESI EDR data, leading to the identification of new cataclysmic variables with follow-up from eROSITA and Keck.

- ² Napolitano, Lucas; Pandey, Agnesh; Myers, Adam D.; Lan, Ting-Wen; **Anand, Abhijeet**; et al., 2023, *Detecting and Characterizing Mg II Absorption in DESI Survey Validation Quasar Spectra*, The Astronomical Journal, **166**, 99 (arXiv:2305.20016) [29 citations]

My role: Contributed to algorithm development and to the physical interpretation of Mg II absorbers detected in DESI early quasar data.

- ¹ Ayromlou, Mohammadreza; Kauffmann, Guinevere; **Anand, Abhijeet**; & White, Simon D. M., 2023, *The physical origin of galactic conformity: from theory to observation*, Monthly Notices of the Royal Astronomical Society, **519**, 1913 (arXiv:2207.02218) [29 citations]

My role: Contributed to the comparison of conformity signals in blue and red galaxies between L-Galaxies simulations and observations from the Legacy Imaging Surveys and SDSS.

Collaboration papers (citations: 6075)

- ³⁶ Dey, Biprateep; Newman, Jeffrey A.; Zhang, Tianqing; Aguilar, J.; Ahlen, S.; et al. (incl. **A. Anand**), 2026, *Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration*, The Astronomical Journal, **172**, 65 (arXiv:2604.06143)
- ³⁵ Moore, Samuel G.; Cole, Shaun; Wilson, Michael; Norberg, Peder; Moustakas, John; et al. (incl. **A. Anand**), 2026, *DESI DR2 Galaxy luminosity functions*, Monthly Notices of the Royal Astronomical Society, **549**, stag866 (arXiv:2511.01803) [6 citations]
- ³⁴ Negi, Vibhore; Joshi, Ravi; Vishnupriya, B.; Wu, Xue-Bing; Yesuf, Hassen M.; et al. (incl. **A. Anand**), 2026, *DESI as sparse Integral Field Spectrograph I: Spatially resolved chemical enrichment in star-forming galaxies at $z \leq 0.1$* , ArXiv, arXiv:2606.12541 (arXiv:2606.12541)
- ³³ DESI Collaboration; Abdul Karim, M.; Adame, A. G.; Aguado, D.; Aguilar, J.; et al. (incl. **A. Anand**), 2026, *Data Release 1 of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **171**, 285 (arXiv:2503.14745) [480 citations]
- ³² Turner, Wynne; Cuceu, Andrei; Martini, Paul; Aguilar, J.; Ahlen, S.; et al. (incl. **A. Anand**), 2026, *Probing the limits of cosmological information from the Lyman- α forest 2-point correlation functions*, Journal of Cosmology and Astroparticle Physics, **2026**, 010 (arXiv:2509.14322) [3 citations]
- ³¹ Zaborowski, E. A.; Taylor, P.; Honscheid, K.; Cuceu, A.; de Mattia, A.; et al. (incl. **A. Anand**),

- 2026, *$H < \text{SUB} > 0$ without the sound horizon (or supernovae): A 2% measurement in DESI DR1*, Journal of Cosmology and Astroparticle Physics, **2026**, 004 (arXiv:2510.19149) [4 citations]
- 30 Karaçaylı, Naim Göksel; Ravoux, Corentin; Martini, Paul; Goff, Jean-Marc Le; Armengaud, Eric; *et al.* (incl. **A. Anand**), 2026, *DESI DR1 Ly α 1D power spectrum: Validation of estimators*, Journal of Cosmology and Astroparticle Physics, **2026**, 048 (arXiv:2509.13593) [5 citations]
- 29 Ebina, H.; White, M.; Raichoor, A.; Dey, Arjun; Schlegel, D.; *et al.* (incl. **A. Anand**), 2026, *Clustering analysis of medium-band selected high-redshift galaxies*, Journal of Cosmology and Astroparticle Physics, **2026**, 019 (arXiv:2509.26467) [3 citations]
- 28 Han, Jiwon Jesse; Chiti, Anirudh; Chen, Kai-Feng; Bechtol, Keith; Bellini, Andrea; *et al.* (incl. **A. Anand**), 2026, *A Path to an All-Sky Survey with Roman*, ArXiv, arXiv:2602.21280 (arXiv:2602.21280)
- 27 Hsu, Yuan-Ming; Huang, Xiaosheng; Storfer, Christopher J.; Inchausti, Jose Carlos; Schlegel, David; *et al.* (incl. **A. Anand**), 2026, *A New Way to Discover Strong Gravitational Lenses: Pairwise Spectroscopic Search from DESI DR1*, The Astrophysical Journal Supplement Series, **282**, 41 (arXiv:2509.16033) [2 citations]
- 26 Kamalinejad, Farshad; Slepian, Zachary; Krolewski, Alex; Greco, Alessandro; Ortolá Leonard, William; *et al.* (incl. **A. Anand**), 2026, *First Detection of the Baryon Acoustic Oscillation (BAO) Feature in the 3-Point Correlation Function of DESI DR1 Luminous Red Galaxies*, ArXiv, arXiv:2602.16134 (arXiv:2602.16134) [1 citations]
- 25 Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Siudek, Malgorzata; Aguilar, J.; *et al.* (incl. **A. Anand**), 2025, *Tracing the Cosmic Evolution of the Cool Circumgalactic Medium of Luminous Red Galaxies with DESI Year 1 Data*, ArXiv, arXiv:2512.03845 (arXiv:2512.03845) [1 citations]
- 24 Lange, Johannes U.; Wells, Alexandra; Hearin, Andrew; Beltz-Mohrmann, Gillian; Leauthaud, Alexie; *et al.* (incl. **A. Anand**), 2025, *Cosmological Constraints from Full-Scale Clustering and Galaxy-Galaxy Lensing with DESI DR1*, ArXiv, arXiv:2512.15962 (arXiv:2512.15962) [7 citations]
- 23 Herrera-Alcantar, Hiram K.; Armengaud, Eric; Yèche, Christophe; Gordon, Calum; Casas, Laura; *et al.* (incl. **A. Anand**), 2025, *The Lyman- α forest from LBGs: First 3D correlation measurement with DESI and prospects for cosmology*, Journal of Cosmology and Astroparticle Physics, **2025**, 053 (arXiv:2507.21852) [4 citations]
- 22 Blanco, Diana; Leauthaud, Alexie; Ulf Lange, Johannes; Wright, Angus H.; Hildebrandt, Hendrik; *et al.* (incl. **A. Anand**), 2025, *The Power of DESI for Photometric Redshift Calibration: A Case Study with KiDS-1000*, ArXiv, arXiv:2512.15964 (arXiv:2512.15964) [3 citations]
- 21 McArthur, Everett; Millon, Martin; Powell, Meredith; Wechsler, Risa H.; Pan, Zhiwei; *et al.* (incl. **A. Anand**), 2025, *Quasars acting as Strong Lenses Found in DESI DR1*, ArXiv, arXiv:2511.02009 (arXiv:2511.02009) [1 citations]
- 20 Uzsoy, Ana Sofía M.; Dey, Arjun; Raichoor, Anand; Finkbeiner, Douglas P.; Ramakrishnan, Vandana; *et al.* (incl. **A. Anand**), 2025, *Effect of local environment on Ly α line profile in DESI/ODIN LAEs*, ArXiv, arXiv:2511.17498 (arXiv:2511.17498) [3 citations]
- 19 Crespi, Andrea; Percival, Will J.; Krolewski, Alex; Bonici, Marco; Zhang, Hanyu; *et al.* (incl. **A. Anand**), 2025, *Baryon fraction from the BAO amplitude: a consistent approach to parameterizing perturbation growth*, ArXiv, arXiv:2511.23459 (arXiv:2511.23459)
- 18 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Alam, S.; Allen, L.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints*,

- Physical Review D, **112**, 083515 ([arXiv:2503.14738](#)) [1222 citations]
- 17 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Allende Prieto, C.; Alves, O.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest*, Physical Review D, **112**, 083514 ([arXiv:2503.14739](#)) [227 citations]
 - 16 Brodzeller, A.; Wolfson, M.; Santos, D. M.; Ho, M.; Tan, T.; *et al.* (incl. **A. Anand**), 2025, *Construction of the damped Ly α absorber catalog for DESI DR2 Ly α BAO*, Physical Review D, **112**, 083510 ([arXiv:2503.14740](#)) [21 citations]
 - 15 Rashkovetskyi, Michael; Eisenstein, Daniel J.; Aguilar, Jessica Nicole; Ahlen, Steven; **Anand, Abhijeet**; *et al.*, 2025, *Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect*, The Open Journal of Astrophysics, **8**, 46033 ([arXiv:2508.20904](#))
 - 14 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 008 ([arXiv:2411.12021](#)) [170 citations]
 - 13 Treiber, Helena; Amon, Alexandra; Wechsler, Risa H.; Manwadkar, Viraj; Myles, Justin; *et al.* (incl. **A. Anand**), 2025, *Dwarf galaxy halo masses from spectroscopic and photometric lensing in DESI and DES*, ArXiv, [arXiv:2509.20434](#) ([arXiv:2509.20434](#)) [2 citations]
 - 12 Cuceu, Andrei; Herrera-Alcantar, Hiram K.; Gordon, Calum; Ramírez-Pérez, César; Armengaud, E.; *et al.* (incl. **A. Anand**), 2025, *DESI DR1 Ly α forest: 3D full-shape analysis and cosmological constraints*, ArXiv, [arXiv:2509.15308](#) ([arXiv:2509.15308](#)) [17 citations]
 - 11 Medina, Gustavo E.; Li, Ting S.; Eadie, Gwendolyn M.; Riley, Alexander H.; Valluri, Monica; *et al.* (incl. **A. Anand**), 2025, *The mass of the Milky Way from outer halo stars measured by DESI DR1*, ArXiv, [arXiv:2508.19351](#) ([arXiv:2508.19351](#)) [3 citations]
 - 10 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements*, Journal of Cosmology and Astroparticle Physics, **2025**, 028 ([arXiv:2411.12022](#)) [313 citations]
 - 9 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics*, Journal of Cosmology and Astroparticle Physics, **2025**, 017 ([arXiv:2411.12020](#)) [122 citations]
 - 8 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 III: baryon acoustic oscillations from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 012 ([arXiv:2404.03000](#)) [509 citations]
 - 7 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*, Journal of Cosmology and Astroparticle Physics, **2025**, 021 ([arXiv:2404.03002](#)) [1657 citations]
 - 6 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest*, Journal of Cosmology and Astroparticle Physics, **2025**, 124 ([arXiv:2404.03001](#)) [364 citations]
 - 5 Ross, A. J.; Aguilar, J.; Ahlen, S.; Alam, S.; **Anand, Abhijeet**; *et al.*, 2025, *The construction of large-scale structure catalogs for the Dark Energy Spectroscopic Instrument*, Journal of Cosmology and Astroparticle Physics, **2025**, 125 ([arXiv:2405.16593](#)) [53 citations]
 - 4 Scholte, Dirk; Saintonge, Amélie; Moustakas, John; Catinella, Barbara; Zou, Hu; *et al.* (incl. **A. Anand**), 2024, *The atomic gas sequence and mass-metallicity relation from dwarfs to massive galaxies*, Monthly Notices of the Royal Astronomical Society, **535**, 2341 ([arXiv:2408.03996](#)) [26 citations]

- ³ DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *The Early Data Release of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **168**, 58 (arXiv:2306.06308) [543 citations]
- ² DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **167**, 62 (arXiv:2306.06307) [293 citations]
- ¹ Han, Jiwon Jesse; Dey, Arjun; Price-Whelan, Adrian M.; Najita, Joan; Schlafly, Edward F.; *et al.* (incl. **A. Anand**), 2023, *NANCY: Next-generation All-sky Near-infrared Community survey*, ArXiv, arXiv:2306.11784 (arXiv:2306.11784) [10 citations]